

README FILE

ABC Technologies Customer Automation System using LangGraph

Project Overview

The **ABC Technologies Customer Automation System** is an AI-powered customer support automation project developed using **LangGraph**, **LangChain**, **OpenAI GPT**, and **SQLite**. The system intelligently classifies customer queries, routes them to the appropriate support agent, retrieves relevant knowledge using Retrieval-Augmented Generation (RAG), performs supervisor validation, supports Human-in-the-Loop (HITL) approval for critical requests, stores conversation history in SQLite, and recalls previous interactions when requested.

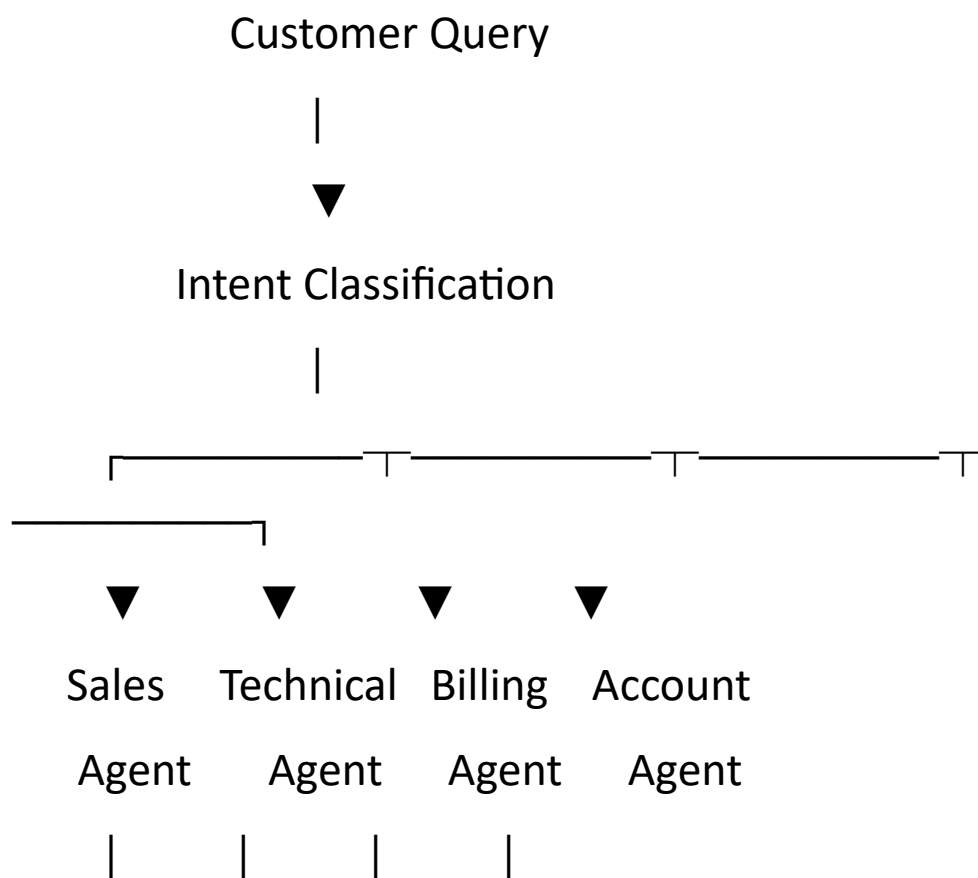
The project demonstrates how multiple AI agents can collaborate in a structured workflow to automate enterprise customer support while maintaining compliance and conversation memory.

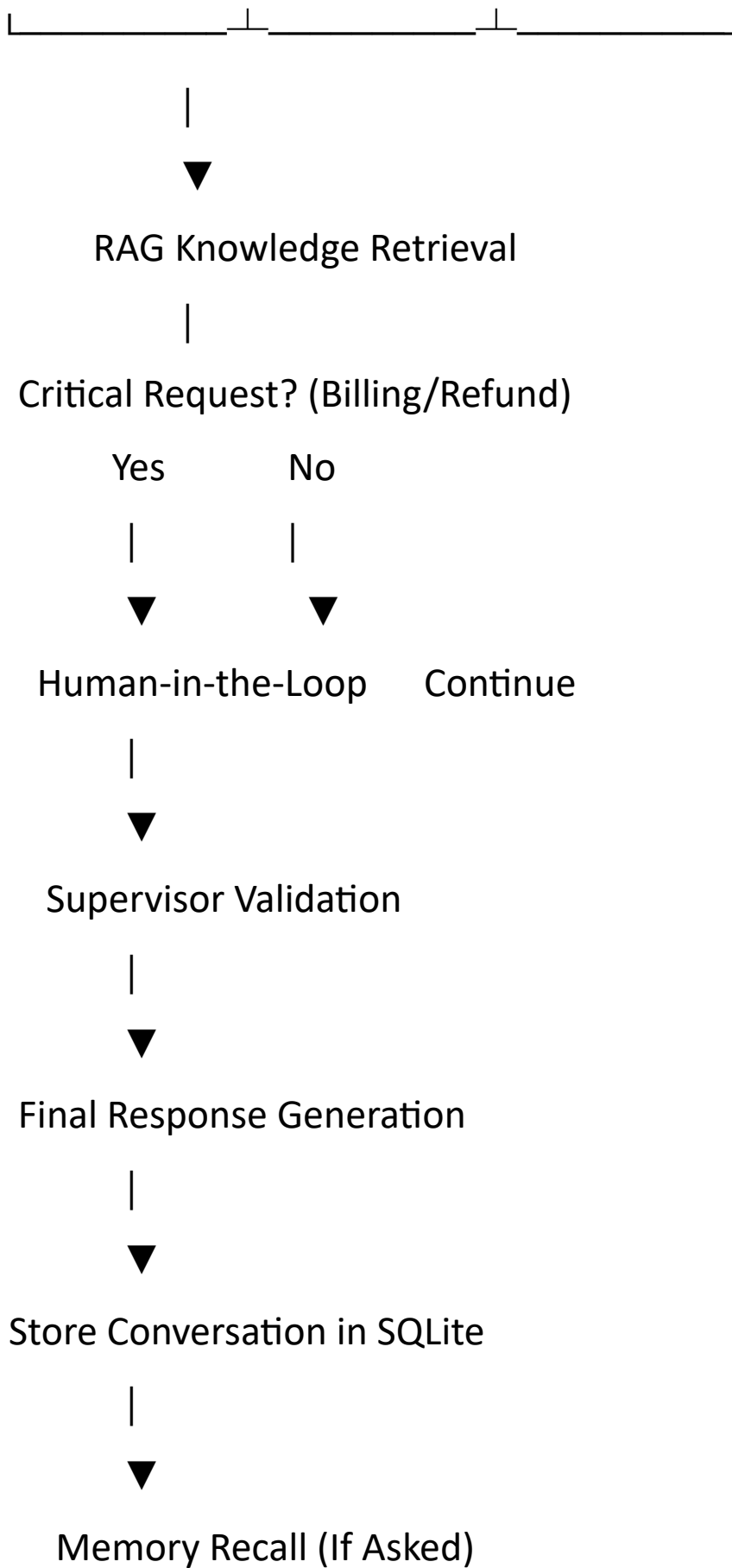
Features

- Intent Classification using AI
- Multi-Agent Routing
 - Sales Agent
 - Technical Support Agent
 - Billing Agent

- Account Support Agent
- Retrieval-Augmented Generation (RAG)
- Human-in-the-Loop (HITL) Approval for Critical Billing Requests
- Supervisor Validation
- SQLite Conversation Memory
- Memory Recall
- Professional Final Response Generation
- Rich Console Output

Project Workflow





Technologies Used

- Python 3.11
- LangGraph
- LangChain
- OpenAI GPT
- SQLite
- Rich
- FAISS / Vector Store (if used)
- Virtual Environment (venv)

Project Structure

ABC_Technologies_Project/

|

└─ solution2.py

└─ support_memory.db

└─ requirements.txt

└─ README.md

└─ workflow_diagram.png

└─ screenshots.pdf

Installation

1. Clone the Repository

```
git clone https://github.com/Ramitha0275/IBM-Assignment-Ramitha-S
```

2. Navigate to the Project Folder

```
cd ABC_Technologies_Project
```

3. Create a Virtual Environment

```
python -m venv venv
```

4. Activate the Virtual Environment

Windows

```
venv\Scripts\activate
```

Linux/macOS

```
source venv/bin/activate
```

5. Install Dependencies

```
pip install -r requirements.txt
```

Running the Project

Execute the following command:

```
python solution2.py
```

Sample Execution

Customer Query 1

Input:

What are the pricing plans available for your software?

Intent:

SALES

RAG:

Pricing Guide Retrieved

Output:

Starter Plan

Professional Plan

Enterprise Plan

Customer Query 2

Input:

I forgot my account password.

Intent:

ACCOUNT

Output:

Password reset instructions generated.

Customer Query 3

Input:

My application crashes whenever I upload a file.

Intent:

TECHNICAL

Output:

Technical troubleshooting steps generated.

Customer Query 4

Input:

I need a refund for my annual subscription.

Intent:

BILLING

Output:

Human approval requested.

Supervisor validation completed.

Refund process initiated.

Customer Query 5

Input:

What was my previous support issue?

Output:

Conversation history retrieved from SQLite memory.

Key Components

Intent Classification

Automatically classifies incoming customer requests into the appropriate support category.

Supported categories include:

- Sales
 - Technical Support
 - Billing
 - Account Support
 - Memory Recall
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RAG Retrieval

Relevant information is retrieved from the company's knowledge base before generating the response, ensuring accurate and context-aware answers.

Human-in-the-Loop (HITL)

Critical billing and refund requests require manual approval before processing, ensuring compliance with company policies.

Supervisor Validation

Every generated response passes through a supervisor validation stage before being returned to the customer.

SQLite Memory

All customer interactions are stored in a SQLite database, enabling conversation history retrieval and contextual responses.

Expected Outputs

The system demonstrates:

- Intent Classification
- Agent Routing
- RAG Context Retrieval
- Human-in-the-Loop Approval
- Supervisor Validation
- Memory Storage

- Memory Recall
 - Final AI Response Generation
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Future Improvements

- Web-based user interface
 - Voice assistant integration
 - Multi-language support
 - Real-time customer analytics dashboard
 - Authentication and user management
 - Cloud deployment using Docker and Kubernetes
-

Author

Name: Your Name

Project: ABC Technologies Customer Automation System using LangGraph

Technology Stack: Python, LangGraph, LangChain, OpenAI GPT, SQLite